

Muhammad Zayan

+92 331 6074205 | muhammadzayan2004@gmail.com | github.com/mzayan-bit
mzayan-bit.github.io | Islamabad, Pakistan

EDUCATION

Ghulam Ishaq Khan Institute (GIKI)

Bachelor of Science in Computer Science

- CGPA: 3.17/4.00

Topi, Pakistan

Sep. 2023 – May 2027

EXPERIENCE

AI Intern

May 17, 2026 – Present

DecodeLabs

Remote

- Developed an AI training pipeline including rule-based chatbots, KNN classification, and TF-IDF engines.
- Implemented data preprocessing (StandardScaler) to normalize datasets and prevent magnitude bias.
- Validated ML outputs using Confusion Matrices and weighted F1 Scores to address dataset imbalances.
- Engineered content-based recommendation logic to mitigate choice overload and solve the Cold Start problem.

AI Intern

June 2, 2025 – July 1, 2025

Integration Xperts

- Engineered computer vision solutions using YOLO and MediaPipe for object detection and pose estimation.
- Optimized ML model inference times for real-time applications within cross-functional teams.
- Preprocessed datasets using Pandas and NumPy to streamline the training of deep learning models.
- Curated and annotated custom image datasets utilizing Roboflow to train robust object detection models.

PROJECTS

VisionTraceAI | *React, FastAPI, Kafka, YOLO11, SigLIP, LangGraph*

- Architected a real-time, event-driven video analytics platform utilizing YOLO11, Kafka, and FastReID for cross-camera tracking.
- Engineered a zero-shot semantic search pipeline using Google's SigLIP and Qdrant for natural language text-to-video retrieval.
- Developed a React and FastAPI full-stack dashboard featuring a LangGraph autonomous agent to investigate spatial-temporal memory.
- [Source Code: github.com/mzayan-bit/VisionTraceAI](https://github.com/mzayan-bit/VisionTraceAI)

AI Gym Vision | *Python, MediaPipe, OpenCV, Linear Algebra*

- Developed a CV fitness assistant to track human pose in real-time utilizing the MediaPipe framework.
- Implemented vector mathematics to calculate joint angles for automated exercise form feedback.
- Collaborated in a team environment to integrate vision models with a user-friendly interface.
- [Source Code: github.com/mzayan-bit/AI-Gym-Vision](https://github.com/mzayan-bit/AI-Gym-Vision)

Matchmaker & Classification Engine | *Python, Scikit-Learn, Pandas, TF-IDF*

- Architected an IPO recommendation engine using TF-IDF vectorization and Cosine Similarity to map user skill profiles to career trajectories.
- Built a K-Nearest Neighbors supervised learning model to accurately classify structured datasets.
- Engineered a deterministic, rule-based chatbot with O(1) dictionary logic and sanitization pipelines.
- [Source Code: github.com/mzayan-bit/Decode_Labs_Internship](https://github.com/mzayan-bit/Decode_Labs_Internship)

RELEVANT COURSEWORK

Core CS: Data Structures & Algorithms, Design and Analysis of Algorithms, OOP, DBMS, OS

Math & Stats: Linear Algebra, Discrete Mathematics, Inferential Statistics & Applied Probability

Specialized: Machine Learning, Flutter App Development, DevOps

TECHNICAL SKILLS

Languages: Python, C++, C, Dart, SQL, Bash/Shell Scripting

Frameworks & Libraries: Django, Flutter, Scikit-learn, MediaPipe, YOLO, OpenCV, Pandas, NumPy

Tools & Platforms: Git, GitHub, GitHub Actions, CI/CD, Docker, Terraform, AWS, Linux, PostgreSQL